Operating Systems Exam Questions

Section 1: Operating System Basics, Structure and Operation

a. The source of an operation system may contain more than ten million lines of code.

b. The kernel runs in protected mode and governs applications operating in user mode.

c. In embedded operating systems the application and the OS are not always separated, sometimes they are compiled into a single executable.

d. Systemd is a sysinit alternative.

e. A modular kernel may not be a monolithic program.

Section 2: Task Management

a. Response time is the time interval between task creation and its first output (response).

b. The ps command on Unix systems lists files.

c. Memory-intensive tasks may become I/O-intensive if there is not enough memory.

d. A process is a sequentially executed thread.

e. All administrative data of a task is stored in the kernel's memory.

Section 3: Scheduling

a. It is very hard to implement the shortest job first (SJF) scheduler in real operating systems.

b. Schedulers schedule processes not threads.

c. The Round-robin scheduling avoids starvation.

d. The SJF may be seen as a priority-based scheduler.

e. Today's operating systems mainly use the hard CPU-affinity in multiprocessor scheduling.

Section 4: Memory Management

a. Tasks may not use more memory that is physically available in the system.

b. The number of page tables are significantly higher than the number of frame tables.

c. If page fault frequency is higher, then memory-intensive tasks spend more time in blocked (waiting) state.

d. The page fault frequency always decreases if we add more physical memory to the system.

e. Today's operating systems mainly use anticipatory paging because it is able to predict what pages will be used in the near future.

Open Questions

1. List a few examples for heterogenous systems. **(2 p)**
2. The message queue is a direct / indirect communication method. (Underline the appropriate) **(1 p)**
3. The main parts of a task's runtime context are ... **(2 p)**
4. Main frame table fields are ... **(2 p)**
5. What is the purpose of the copy-on-write technique? **(1 p)**
6. What is the main difference between the clock and the second chance page replacement algorithms? **(1 p)**
7. Write down the names of two different command line shells. **(8 p)**
8. What is the role of the rpcgen command in remote procedure call implementations? **(1 p)**
9. Write down a small code fragment that illustrates how semaphore operations can protect a critical section! **(2 p)**
10. What is more reliable: RAID0 or RAID1? **(1 p)**